

# VICTOR IBHAFIDON

## Research Engineer, Machine Learning

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### PROFESSIONAL SUMMARY

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My approach to software engineering is grounded in the belief that good software is built through thoughtful design, sound engineering, and continuous iteration. I've worked on AI/ML solutions across software engineering, data, security, and cloud, gaining experience through building, iterating, and learning from challenges along the way, while balancing technical requirements with broader business needs. My philosophy is simple: I don't just want something that works in a demo. I want to understand what can go wrong, prove it works, and build it properly.

### TECHNICAL SKILLS

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**AI & Machine Learning:** LLMs, Generative AI, AI Agents, Multi-Agent Systems, LangGraph, MCP, JSON-RPC, Tool Calling, RAG, GraphRAG, Prompt Engineering, AI Evaluation, PyTorch, Transformers, Scikit-learn, Computer Vision, NLP

**MLOps & Model Operations:** MLflow, Model Training, Model Evaluation, Model Monitoring, Drift Detection, Experiment Tracking, Model Deployment

**Backend & Data:** FastAPI, Django, REST APIs, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, Redis, Neo4j, pgvector, Vector Search, Data Pipelines, ETL

**Programming:** Python, TypeScript, JavaScript, SQL, Bash

**Security & AI Governance:** AI Security, AI Governance, Authentication, Authorization, RBAC, API Security, OWASP Top 10, Prompt Injection Mitigation, AI Safety, Responsible AI, GDPR

**Cloud & DevOps:** AWS (EC2, S3, Lambda, IAM, EKS), Microsoft Azure, Azure OpenAI, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Nginx, OpenTelemetry, Prometheus, Grafana

### PROFESSIONAL EXPERIENCE

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#### Software Engineer (AI/ML), Bastion Shield Technologies

London, United Kingdom | April 2026 – Present

- Developed behavioural anomaly detection combining statistical methods with usage forecasting to identify unusual agent activity. The system achieved a 0.981 ROC-AUC against an Isolation Forest baseline and was integrated into the TypeScript runtime for real-time scoring.
- Engineered the backend in Python, TypeScript, and PostgreSQL, implementing RBAC, server-side tenant isolation, secure authentication, API-key management, and an SDK for external teams to register agents and stream policy decisions. Designed tamper-evident audit logging with offline verification, strengthened data privacy controls, and remediated four production vulnerabilities, including a session-token forgery path.
- Built Bulwark, a multi-tenant AI agent governance platform that gives organisations control and visibility over AI agents. Designed and implemented agent identity and lifecycle management, policy enforcement, human oversight workflows, and detection of unregistered agents, orphaned ownership, and stale credentials.
- Owned the cloud infrastructure using Terraform-managed AWS environments across development, staging, and production. Containerised the platform with Docker and delivered it through GitHub Actions CI/CD, while working on distributed system patterns including queues, event streaming, caching, and horizontal scaling to improve performance and latency.

#### Full Stack Software Engineer, Basketball Nxtion

London, United Kingdom | February 2026 – April 2026

- Designed the PostgreSQL data architecture and analytics pipelines for the MVP, structuring user, activity, engagement, and platform data to support data ingestion, transformation, analysis, and business intelligence. Built data workflows enabling insights into platform performance, user behaviour, engagement, and product adoption.
- Designed and integrated an AI architecture connecting LLM/ML services with PostgreSQL data, backend APIs, and application business logic. Built workflows for personalised recommendations, intelligent user matching, and content discovery, using user and platform data to generate relevant results, improve engagement, and automate discovery workflows.
- Developed the MVP of Plixground, a mobile application connecting basketball players, creators, and sports communities, contributing across product design, application architecture, backend development, APIs, authentication, and user workflows.
- Deployed and maintained the MVP infrastructure on AWS, implementing Docker, CI/CD, testing, rollbacks, monitoring, and third-party integrations while collaborating with frontend engineers and stakeholders throughout the software development lifecycle from requirements and development through testing, deployment, and iteration.

## Full Stack Python Engineer Intern, CyberAI Technologies

London, United Kingdom | September 2025 – November 2025

- Contributed to the development of RAG and conversational AI workflows using Python and Django, working on document ingestion, chunking, embeddings, vector-based retrieval, prompt construction, conversation context, REST APIs, and LLM integrations while learning how knowledge-based AI systems are designed and evaluated.
- Developed backend functionality and AI workflow improvements, working on API development, debugging, testing, performance optimisation, knowledge retrieval, and integration between AI services and the Django application.
- Collaborated with senior AI engineers to develop Surfyn AI, a conversational AI customer support platform enabling businesses to deploy knowledge-based AI assistants grounded in websites and documents to automate customer interactions and provide relevant, context-aware responses.
- Worked with the AI Security Engineering team on AI evaluation, testing, safety, and deployment workflows, helping investigate retrieval quality, response relevance, hallucination risks, prompt injection, sensitive data handling, logging, and application reliability across production AI systems.

## SELECTED PROJECTS

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**Autonomous Energy Grid & Smart Battery Management:** Engineered an end-to-end ML/MLOps system to support battery optimisation and energy-management decisions, transforming raw energy and battery data into features used for predictive modelling and anomaly detection. Evaluated and iterated on candidate models using validation performance and operational requirements, tuning model parameters to improve predictive performance while considering the trade-off between accuracy, inference cost, and reliability. Built the training and deployment pipeline with MLflow, Docker, Kubernetes, and GitHub Actions, enabling reproducible experiments, versioned models, automated deployment, and consistent promotion of models from development into serving environments. Implemented Prometheus, Grafana, and OpenTelemetry monitoring with drift detection to identify changes in data and model behaviour after deployment, allowing model performance and reliability to be tracked over time. GitHub:

[github.com/Web8080/Energy-Grid-and-Battery-Management-System](https://github.com/Web8080/Energy-Grid-and-Battery-Management-System)

**Medical Imaging AI:** Developed a supervised computer vision system for classifying 231,444 medical images, designing the data preparation and augmentation pipeline to improve the quality and consistency of training data and reduce sensitivity to variations in the input images. Iteratively trained and evaluated PyTorch CNN models, analysing classification metrics and failure patterns to guide model and preprocessing changes, achieving up to 92% classification accuracy. Converted the trained model into a deployable inference service using Docker and REST APIs, separating model training from inference so models could be versioned and deployed independently. Added health checks, audit trails, and runtime monitoring to make model serving more observable and reliable beyond the training environment. GitHub:

[github.com/Web8080/Medical\\_Imaging\\_AI\\_API\\_Research\\_Paper\\_and\\_web\\_app](https://github.com/Web8080/Medical_Imaging_AI_API_Research_Paper_and_web_app)

## EDUCATION

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**MSc Data Science**, Nottingham Trent University, United Kingdom (2023 – 2024)

**BSc Computer Science**, Ambrose Alli University, Nigeria (2016 – 2021)